

Project Proposal: Space Debris Generation and Orbital Propagation Prediction Using AI/ML

Project Title: Space Debris Generation and Orbital Propagation Prediction Using AI/ML

Student Team

1. Vivek Aggarwal – S4015465
2. Truc
3. Ashwin
4. Ravindu

Industrial Supervisor: Dr. Abhiram Jayan Nair (Founder, CEO, CTO, ERETS)

Academic Supervisor: Dr. Ke Deng

Date: 25th Aug '24

Partner/Client and Background

Partner Background:

Partner: ERETS (European Research Establishment for the Sustainable Space)

Our client for this project is ERETS Space, a private organization dedicated to space research with their HQ in London, UK. A multidisciplinary team of engineers specializing in their respective fields come together as ERETS to integrate AI, optics, autonomous robotics, and advanced propulsion systems to tackle one of the most pressing challenges in space—space debris. The company is developing technologies aimed at eliminating space debris from Low Earth Orbit (LEO), and aims to de-orbit objects measuring 1cm – 10cm, in order for it to burn through our atmosphere. The organization's mission takes humanity one step closer towards sustainable space exploration, which happens to be a critical concern in the New Space age.

Business Objectives

NASA defines space debris as "any man-made object in orbit about the Earth which no longer serves a useful function" (2020, cited in Miller, 2021). The origin of this space debris can come from various sources; it can result from the sudden breakup of a satellite in orbit or an accidental collision. However, the biggest source of space debris comes from anti-satellite tests (Miller, 2021). The United States Space Force stated that due to the increasing amount of space debris, debris-related collisions will become more widespread, which will further affect orbital flight safety (2020, cited in Miller, 2021). In fact, the fact that space debris represents a significant risk to satellite operation was also mentioned in research by Lewis, Swinerd, and Newland (2011), which means that over the last 10 years, there have not been many changes made to address the problem. According to the European Space Agency (ESA), there are more than 29,000 pieces of space debris larger than 10 cm, 670,000 larger than 1 cm, and more than 170 million larger than 1 mm. These facts are strong evidence that this is a growing and urgent problem.

The main objective of the business is in the market of space debris management, ranging from space debris mitigation to tracking and predictions for events using DS/AI techniques. These operations will be carried out by two of their engineered satellites: ERETS-C satellite

and ERETS-D. ERETS-C is a satellite constellation specified for debris tracking and detection in Low Earth Orbit (LEO) and Medium Earth Orbit (MEO)

AI and DS Application

ERETS Space believes that Data Science (DS) and Artificial Intelligence (AI) can help solve their problems for the following reason: Traditional prediction methods, reliant on historical data and physical modelling, often fall short in accurately forecasting debris generation and its potential impacts. AI and ML techniques can predict space debris generation more effectively based on real-time debris generation events recorded in past decades (like anti-satellite missile tests or debris and satellite collisions).

Project Aims

This project will be divided into 2 stages, each having its own objective. For the first stage of the project, our aim is to prepare the dataset for the development of our model by:

1. **Creating a Learning Database:** Develop a learning database of space debris propagation, ie orbital parameters and decay rates, based on historical and real-time data
2. **Developing a Predictive Pipeline:** Implement a machine learning pipeline to predict space debris decay rates, re-entry parameters, and potential collision events.

Extended Aim: Feed the model with real-time data for adaptive learning, in order to achieve dynamic predictions in real time, with minimal human interference.

Methodology and Expected Deliverables

Methodology:

1. **Data Collection and Preprocessing:**

- Task 1: Web scraping debris data from Space-Track.org to create a database of orbital parameters of debris.
 - Task 2: Designing mathematical models to calculate debris propagation and re-entry locations.
2. Model Development:
 - Task 3: Develop a predictive pipeline using historical collision events (e.g., 2009 Kosmos-Iridium collision) to estimate debris behavior, which includes debris cloud generation, post collision propagation and energies, sub-collisions, etc.
 3. Model Optimization:
 - Task 4: Optimize automated real-time data integration into the model and implement adaptive learning mechanisms to keep the model self-aware of dynamic space conditions.
 4. Evaluation and Testing:
 - Task 5: Test and validate the model using simulated scenarios.

Expected Deliverables:

- Learning Database
- Predictive Pipeline
- Final Report
- Presentation

Project Plan

Week-by-Week Timeline:

5. **12 Aug - 18 Aug:**
 - Finalize project proposal.
 - Begin feature engineering and ML model selection.
 - **Deliverable:** Project Proposal Report and Presentation (University Assessment 1).
6. **19 Aug - 25 Aug:**
 - Initial meeting with industry and academic supervisors.
 - **Deliverable:** Background study report submission (ERETS 1st progress report).
7. **26 Aug - 01 Sep:**
 - Continue working on data collection and preprocessing.
 - **Deliverable:** Submission of Module 1.
8. **2 Sep - 6 Sep:**
 - Semester Break.
9. **9 Sep - 15 Sep:**
 - Develop and test the mathematical model for debris decay.
 - **Deliverable:** Submission of Module 2.
10. **16 Sep - 29 Sep:**
 - Develop predictive pipeline based on historical data.
 - **Deliverable:** Submission of Module 4.

11. 30 Sep - 6 Oct:

- Begin working on the final report and presentation.
- **Deliverable:** Submission of Module 5 (Final Feedback report to RMIT).

12. 7 Oct - 27 Oct:

- Finalize the model and report.
- **Deliverable:** Submission of Module 6 with final report and presentation slides.

13. 28 Oct - 3 Nov:

- Present the project outcomes.
- **Deliverable:** Oral project presentation and code package handover (Project Showcase).

Roles and Responsibilities:

TBD

Conflict Resolution:

- Weekly meetings with the team and supervisors to resolve doubts
- Whatsapp to be used as an unofficial way of communication within the team
- Unresolved issues will be escalated to higher authority.

References

Lewis, H.G., Swinerd, G.G. and Newland, R.J., 2011. The space debris environment: future evolution. The Aeronautical Journal, 115(1166), pp.241-247.

Miller, G.D., 2021. Deterrence by debris: the downside to cleaning up space. Space Policy, 58, p.101447.